

Four-Core Pipelined Implied Volatility & Greeks Engine (Artix-7 200T @ 100 MHz)

Stage 1: Ingress Registration & Queuing (1 Cycle)

- AXI4-Stream 256-bit Ingress: {S, K, T, r, C_mkt, TID}
- Single-Cycle Register + TID Tag
 - No Ingress Divider (Zero Division Latency)
- Q8.24 Fixed-Point Precision

Zero-BRAM Context Store

- 64 Context Slots / Core (256 Total)
- 60 Active + 4 Headroom Invariant (Prevents Recirculation Deadlock)
 - Distributed LUTRAM & SRL32
- 0 Block RAM / UltraRAM in Core
- Leaves 365 BRAM36 Free for System

Stage 2: Analytical Seeder (Brenner-Subrahmanyam, 64 Cyc)

$$\sigma_0 \approx \frac{2,5066 C_{mkt}}{\sqrt{T} (S + K)/2}$$

- Shared Digit-Recurrence $\sqrt{T}$ (29 Cyc, II = 1)
 - Forwards $\sqrt{T}$ Directly to Stage 3
- Non-Restoring Divider (33 Cyc, II = 1)
- Analytical Initial Estimate (Sec 2.4):
Med Abs Err: 0.1037 vol (1,037 vol-bps) | Rel: 30.52% (11.53%)

Stage 3: Core Feedforward Datapath (126 Cycles, II = 1)

- II = 1 Pipeline (Reused Across Passes)
- Scale-Invariant Padé Ratio (S-K)/(S+K)
- 18-Cyc CORDIC Log Fallback (SRL15-Matched)
 - Horner 5th-Order Poly CDF N(d1), N(d2)
- Hyperbolic CORDIC for exp(-rT) & exp(-d1^2/2)

Stage 4: NR Update & Dividers (33 Cycles, II = 1)

- Convergence: $|C_{BS} - C_{mkt}| \leq \0.01
- **u_nr_divider**: $\Delta\sigma = (C_{BS} - C_{mkt})/\nu$
- **u_gamma_divider**: $\Gamma = \phi(d_1)/(S\sigma\sqrt{T})$
- Greeks: $\Delta = N(d_1)$, $\nu = S\sqrt{T}\phi(d_1)$
- TID-Scoreboard Priority Loopback

Stage 5: Multi-Core Arbitrer & 128-Bit Egress (2 Cycles)

- Round-Robin 4-Core Drain Arbitrer
 - 128-Bit Data Payload:
 - [127:122] Local TID (6b)
 - [121:96] Delta Greek (26b, Q2.24)
 - [95:64] Gamma Greek (32b, Q8.24)*
 - [63:32] Vega Greek (32b, Q8.24)*
 - [31:0] sigma Implied Vol (32b, Q8.24)
 - Sideband: Core ID (2b)
- *Carries ν^* , Γ^* for normalized inputs

Nonconverged $\rightarrow \sigma_{n+1}$ Recirculation (Priority Restore to Stage 3)

CBS, ν (126 Cyc)

$\sigma_0, \sqrt{T}$ (Forward)

Converged $\rightarrow$ Egress

AXI4-Stream Egress (128b Payload + 2b Core ID Sideband)